

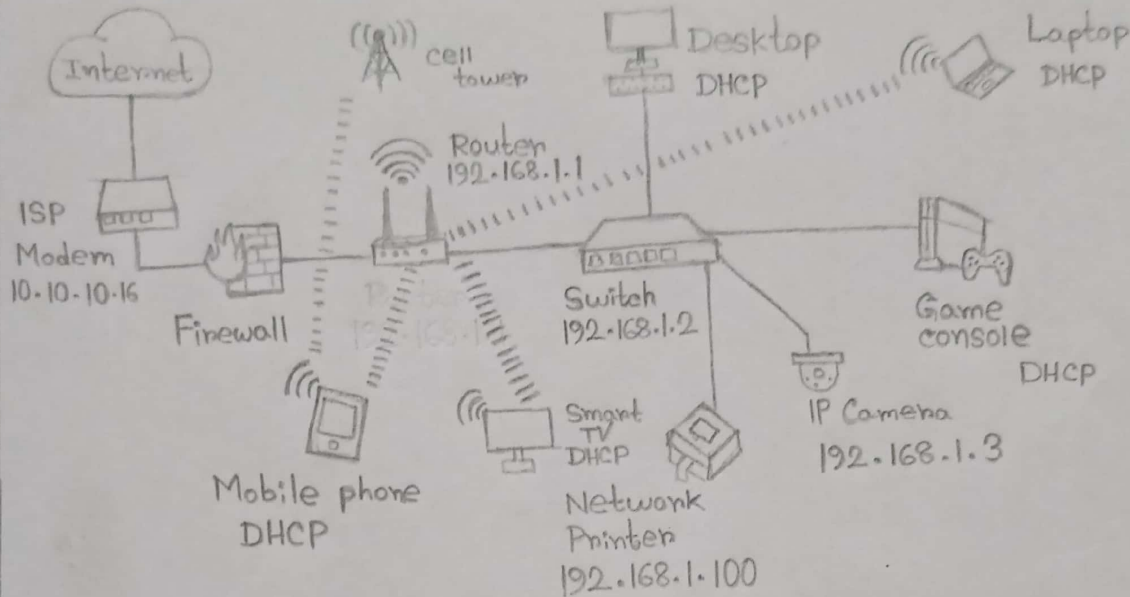
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Subject: CA - Network (CA-N)

Assignment 1: Draw your Home Network Topology and explain it.

⇒ My home network uses a hybrid star topology, where all devices connect to a central router, which acts as the central hub for all communication, some via network switch and smartphones have an additional 4G/5G connection option.



- **Wired connections:** Ethernet (CAT5/CAT6) cables connect high-speed devices.
- **Wireless connections:** Wi-Fi connects mobile devices and they can switch to 4G/5G cell tower when outside or Wi-Fi is unavailable.
- **Firewall:** Protects against unauthorized access.
- **Switch:** Extends wired connectivity.

□ ISP Modem (Wide Area Network - WAN Gateway)

- **Function:** Converts the ISP's fiber/cable signal into a usable internet connection.
- **Technical Aspects:**
 - **Public IP Address:** Assigned by the ISP.
 - **DHCP:** Assigns an IP dynamically.
 - **NAT:** Translates private home IPs to public IP.
- It is connected to Firewall via CAT6 Ethernet cable.

❑ Firewall (For Security)

- Function: Filters incoming/outgoing traffic, preventing cyber threats.
- Technical Aspects:
 - Packet Filtering: Inspects incoming packets.
 - Intrusion Detecting System & Intrusion Prevention System.
 - Port Forwarding: Allows remote access to specific devices.
- It is connected to Router via CAT6 cable.

❑ Wi-Fi Router (Local Area Network - LAN Router)

- Function: Manages wired/wireless connections, assigns IPs, and routes traffic.
- Technical Aspects:
 - Wi-Fi Standards: 802.11ac, 802.11ax (Wi-Fi 6).
 - SSID: The network name
 - WPA3 Encryption: Secures Wi-Fi connections
 - Dual Band (2.4GHz & 5GHz): Optimized performance, 2.4GHz provides longer range but less speed and 5GHz provides shorter range but more speed.
 - DHCP Server: Assigns private IP addresses to devices.
 - Routing Table: Directs network traffic.
- It is connected to switch via CAT6 cable.

❑ Network Switch (Wired connectivity Hub)

- Function: Expands Ethernet connections for wired devices.
- Technical Aspects:
 - Layer 2 switching (MAC Address-Based): Sends data to correct device.
 - Full-Duplex Communication: Simultaneous sending & receiving of data.
 - PoE (Power over Ethernet): Powers IP cameras via Ethernet
 - VLANs (Virtual LANs): Can segment the network for security.

- Switch connected to Router via CAT6 cable.

□ End Devices (LAN Clients)

• Wired Devices

- PC, TV, Laptop (Optional)

- IP Camera

- Uses PoE (Power over Ethernet) or Wi-Fi

- Supports RTSP (Real-Time Streaming Protocol) for remote viewing.

• Wireless Device

- Smartphones (Dual Connectivity), Laptop, TV

- Smartphone can use Wi-Fi and when Wi-Fi is not available it can use 4G/5G cellular data.

- Wired devices connect via CAT6, wireless devices via Wi-Fi

□ Network Communication

▣ Internet Access Flow

- Request : PC → Router → Firewall → ISP Modem → Internet

- Response : Internet → ISP Modem → Firewall → Router → PC

▣ Local Device Communication

- Wired devices talk via the network switch.

- Wireless devices talk via the Wi-Fi routers.

▣ IP Camera Remote Access (Using Port Forwarding)

- External device requests camera feed using its public IP & port.

- The firewall and router direct the request to the camera's local IP.

□ Key Features of the Network

- Dynamic IP Assignment : The router uses DHCP (Dynamic Host configuration Protocol) to assign IP addresses to devices.

- Network Address Translation (NAT) : The router translates private IP addresses to a public IP for internet access.

- WiFi Bands : Supports 2.4 GHz (longer range, slower speed) and 5 GHz (shorter range, faster speed).
- Security : The router has a built-in firewall and encryption (WPA2/WPA3) to protect against unauthorized access.
- CAT6 Cables : Supports speeds upto 10 Gbps.

□ Advantages of the Network :

- High-Speed Internet Access - Efficient connectivity for all devices via wired (Ethernet) and wireless (Wi-Fi).
- Enhanced Security - The firewall and network segmentation (VLANs) improve protection against cyber threats.
- Scalability - Easily expandable with more devices.

□ Disadvantages of the Network :

- Single Point of Failure - If the router, switch, or firewall fails, the entire network may be disrupted.
- Limited wireless Range - The Wi-Fi signal might not cover large areas without additional access points.

This home network topology effectively balances security, performance, and scalability, ensuring reliable connectivity for various devices. By integrating a firewall and VLANs the network remains secure and efficient. And we can further expand and enhance the network.